

SanTube

JUST DOWNLOAD

A native Android app that downloads videos and music from YouTube, Instagram, Facebook, Reddit and most websites - and plays everything you saved offline.

Version 1.1 | Android 7.0 and newer | Kotlin + Jetpack Compose | No account, no server

Project overview

What it is · what it does · the engine · Magic · performance

SanStudio · Santhosh · October 2026

AT A GLANCE

What is SanTube?

SanTube is a native Android app, written in Kotlin with Jetpack Compose. You share or paste a link from YouTube, Instagram, Facebook, Reddit or almost any website, pick a format, and SanTube downloads it in the background. Everything you save plays offline in a Reels-style feed, and a personal discovery feed called **Magic** finds new videos for you.

There is no account and no SanTube server: the app talks only to the sites you use, plus GitHub to keep its download engine up to date.

7
screens
Home, Reels, Downloads, Share, History, Settings, Magic

13
Magic sources
PeerTube, Internet Archive, YouTube, Vimeo, Bluesky, your own websites...

1.7×
faster MP3
a 1 h 35 min track: 87.9 s → 50.4 s on a real phone

86 MB
APK per CPU type
down from 333 MB for the all-in-one build

86
automated tests
plus testing on two real phones

0
accounts or servers
your downloads and your taste stay on your phone

Contents

1	What it does	the five steps from a link to an offline video
2	Features	downloading, watching offline, discovering, comfort
3	The engine	yt-dlp, FFmpeg, aria2c and how SanTube drives them
4	Fast MP3	encoding on four cores with joins you cannot hear
5	Magic	the personal feed: sources, SanAlgorithm, SanViral
6	Performance	measured on real phones
7	What was done in 1.1	problems found and how they were fixed
8	Quality, limits and tech	testing, honest drawbacks, the tech stack

1 · WHAT IT DOES

From a link to an offline video

Using SanTube takes five steps. The hard parts - understanding each site, picking streams, joining audio and video, converting to MP3 - happen inside the app.

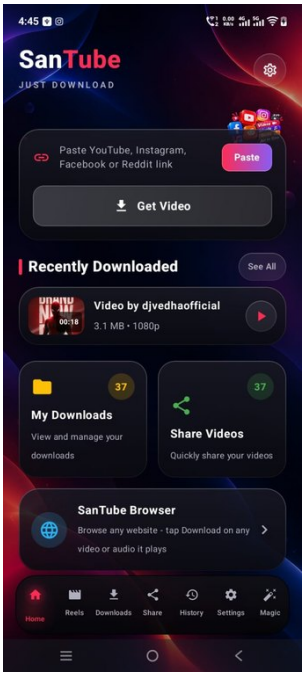
1
Share a link
From Instagram, YouTube, any app's Share menu - or paste one. A link you just copied is offered too.

2
Analyze
SanTube reads the title, thumbnail and every format the site offers, in about 5 seconds.

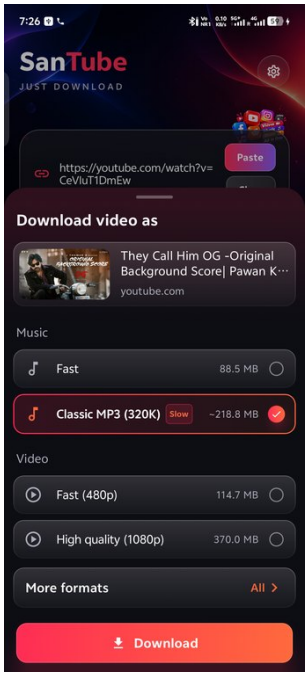
3
Choose
Music: Fast M4A or Classic MP3. Video: 144p up to 1080p and more. Or a Bundle.

4
Download
In the background, even with the app closed. Pause, resume, cancel from the notification.

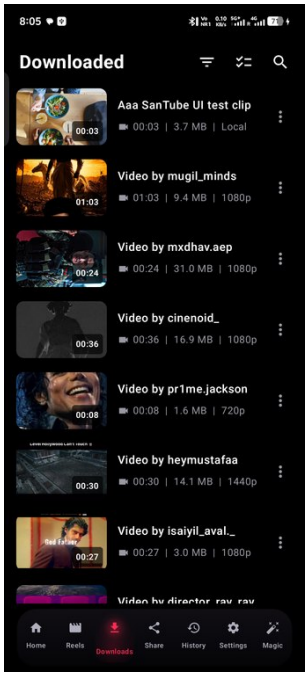
5
Watch offline
In Reels, Downloads, Share and History - and in Gallery and music apps.



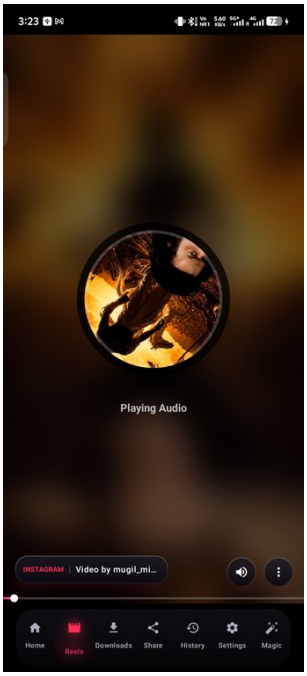
Home: paste a link, recent downloads, Browser



"Download video as" sheet

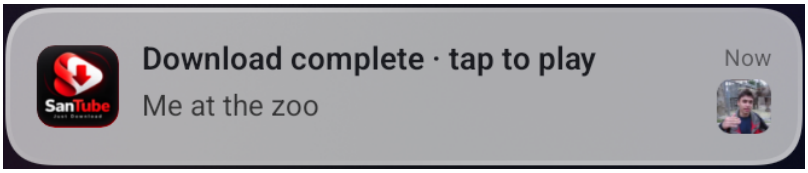


Downloads: everything saved



Reels: offline feed, here playing audio

Finished downloads show up straight away in the phone's Gallery, WhatsApp, file managers and music apps, and the "Download complete" notification shows the video's thumbnail:



2 · FEATURES

What SanTube can do

Download

- **Any link**: YouTube, Instagram, Facebook, Reddit, most other websites, or a direct file link.
- **Music**: Fast (the site's own M4A, no conversion) or Classic MP3 from 70K to 320K.
- **Video**: every quality the site has, with sizes. **Bundle**: video, music and thumbnail in one folder.
- **Background downloads** with Pause / Resume / Cancel; resume continues from the same byte.
- A lost connection waits and continues by itself.
- MP3s carry title, artist, date, cover art and the video's chapters.

Watch and manage offline

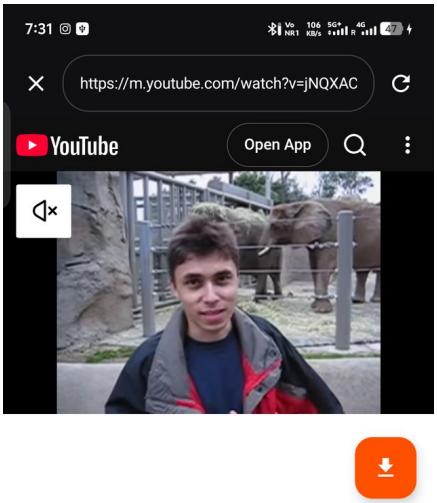
- **Reels**: full-screen vertical feed of all downloads - background play, 2× on long-press, double-tap to skip, landscape full screen, platform filter.
- Files the phone cannot decode play through the built-in **libVLC** engine.
- **Downloads**: search, select, rename, delete. **Share**: open the folder, play inside the app, send.
- **History**: search, filters and a detailed info view (format, size, resolution, codecs, link).
- **Private Safe** behind a password; **hide** downloads from Gallery and music apps.

Discover

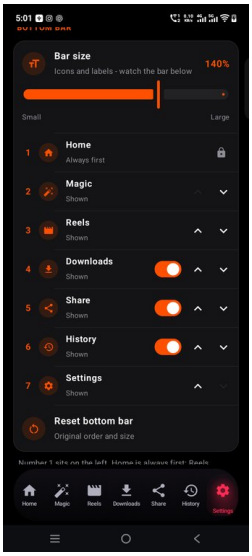
- **Magic**: a personal video feed from 13 sources, including websites you add yourself.
- **SanAlgorithm** learns what you like - on the phone only. **SanViral** puts the best videos first.
- **SanTube Browser**: browse any site and tap Download on what it plays.
- Built-in ad and tracker blocker (Shields) and private browsing.

Comfort

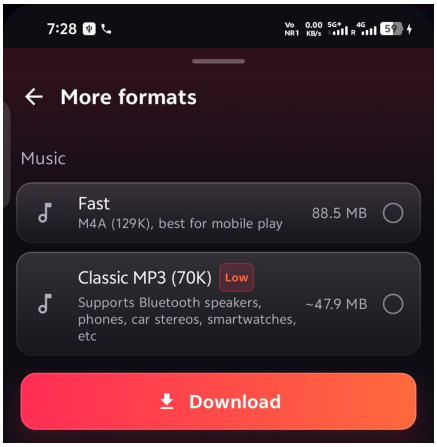
- **One window**: links shared from different apps all go to the SanTube that is already open.
- **Back goes Home first**, then leaves the app - like a normal app.
- **Adjustable bottom bar**: size 80-170 %, tab order (Home first) and which tabs show.
- Works in the phone's **small floating window**: the Download button always stays visible.
- Red-and-white neon “glass” design, dark mode.



SanTube Browser with its Download button



Bottom bar: size and order (Magic moved to 2nd)

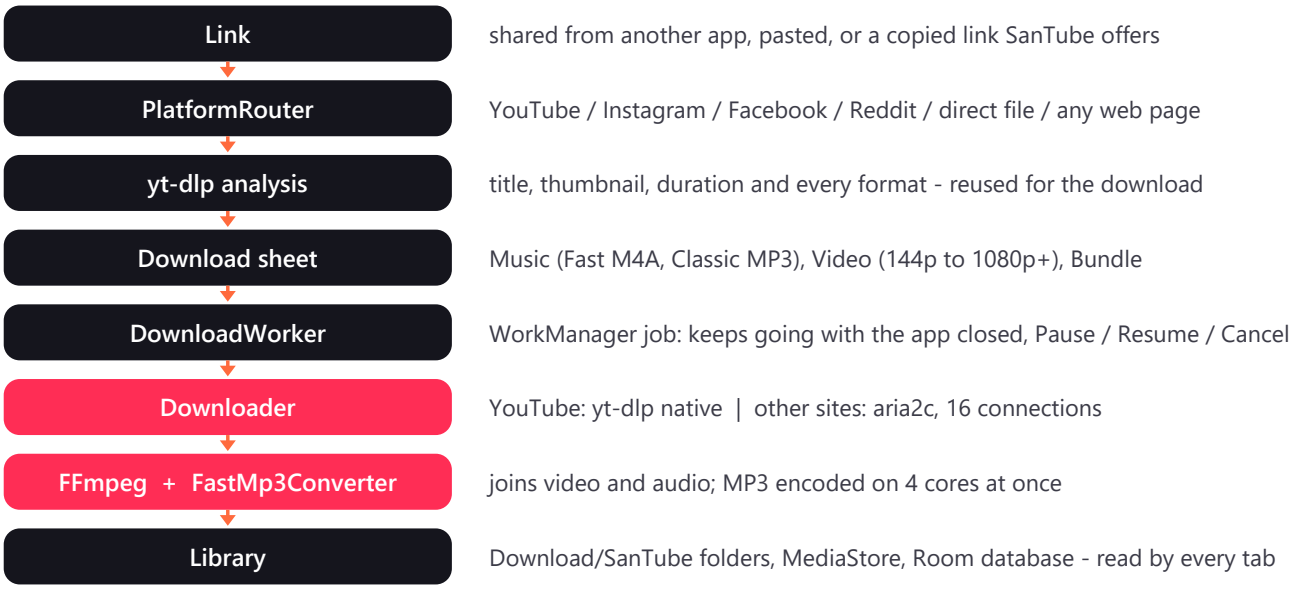


Floating window: list scrolls, Download stays

3 · THE ENGINE

How a link becomes a file

SanTube's download engine is **yt-dlp**, the open-source tool that understands hundreds of video sites. It runs inside the app through **youtubedl-android**, which ships a real Python runtime, **FFmpeg** and **aria2c** as native libraries. SanTube's own code is everything around it: choosing the right streams and downloader, the download queue, notifications, files and the library.



Part	What it does in SanTube
yt-dlp 2026.10.26	Reads each site and finds the real media streams. Shipped inside the APK and updated from GitHub.
Python runtime	Runs yt-dlp on the phone (youtubedl-android 0.18.1), unpacked on the first start.
FFmpeg 7.1 + LAME	Joins separate video and audio streams; encodes MP3; writes tags, cover art and chapters.
aria2c	Downloads from ordinary file servers over 16 connections at once.
WorkManager	Runs each download as a background job that survives the app being closed.
Room + MediaStore	One library for every tab; finished files are registered so other apps see them at once.
Media3 + libVLC	Plays downloads in Reels; libVLC takes over for files the phone's decoder cannot play.

Engine decisions

- **The right downloader per site.** YouTube slows down parallel connections, so YouTube uses yt-dlp's own downloader (a 27 MB video in 1.5 s); every other site uses aria2c (100 MB in 19 s instead of 3 min).
- **Analyze once.** The analysis a download starts from is reused, so the site is not asked twice.
- **Keeps itself current.** The APK carries a recent yt-dlp, checks GitHub once a day - also when opened from a share - and when a known site suddenly fails it updates yt-dlp and tries again once.
- **One shared state.** The app, the notification buttons and the background job read the same task store, so they never disagree about whether a download is paused or running.
- **Lean release build.** R8 shrinks the code, keep rules protect the parts used by reflection, and one APK per CPU type brings the app from 333 MB to 86 MB.

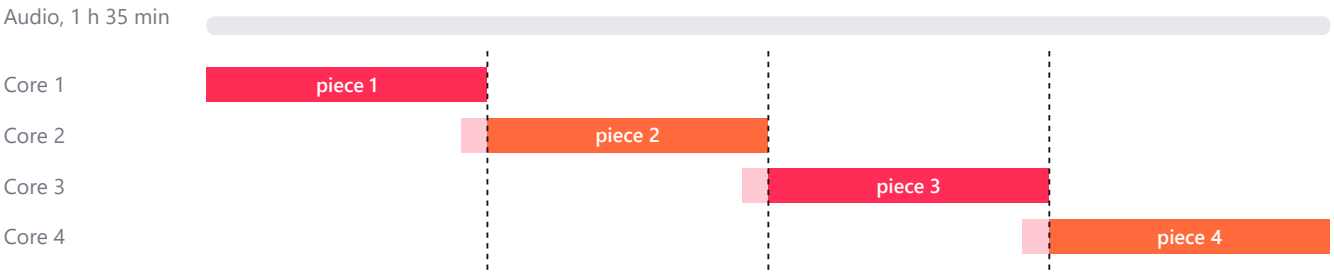
4 · FAST MP3

MP3 on four cores at once

Turning a long video's audio into MP3 used to be SanTube's slowest step: LAME encodes on **one** CPU core while a modern phone has eight, and the file was then rewritten twice more for tags and cover.

	Before	Snaptube (analyzed with REA)	SanTube 1.1
Encoder	LAME on 1 core (FFmpeg)	Shine (fast, lower quality) + FAAD2, 2 threads	LAME on 4 cores at once
Source audio	YouTube Opus	AAC (M4A) only	YouTube Opus (higher quality)
Tags and cover	2 full rewrites	own native code	1 quick copy pass
1 h 35 min track	71 s to convert	not measured	39 s (about 31 s on a cool phone)

How it works



- The audio is cut into four pieces that LAME encodes **at the same time** on four cores.
- Each piece starts about a second early (light red) on the same 1152-sample MP3 frame grid, and LAME's bit reservoir is off, so no frame depends on bytes from the piece before.
- The early frames are dropped and the pieces are joined exactly on frame boundaries.
- Cutting must be exact to the sample. FFmpeg's seek is not, in most formats - a first version put one piece 8 ms late. Now YouTube's Opus is first repackaged into Ogg (exact positions, ~1.7 s) and other formats are cut by counting samples.
- If anything goes wrong, SanTube falls back to a normal one-piece encode, so a download never fails because of it.

Proof

Test on the phone (4 pieces vs. one normal pass)	Result
95 min, MP3 source	byte-identical file (same MD5)
95 min, Opus repackaged into Ogg (what YouTube downloads use)	byte-identical file (same MD5)
20 min, Opus in WebM and AAC in M4A, cut by sample count	byte-identical files
Gluing four plain pieces together, for comparison	an audible click at each join (-0.4 dB peak)

5 · MAGIC

A personal video feed

Magic is SanTube's discovery tab: a full-screen feed of videos chosen for you from free and public sources, with a Download button on every video. You pick topics and sources once; Magic does the rest.

Sources	Which
Free, no key	PeerTube, Internet Archive, Wikimedia Commons, media.ccc.de, Mastodon, Bluesky, Vimeo, Instagram, Facebook
Free API key	YouTube, Dailymotion, Pixabay - skipped when no key is set
Your own	“My websites”: any site you add; SanTube finds its videos itself

SanAlgorithm - learns on the phone

- Learns from how you watch: how long, finished or skipped, came back, downloaded.
- Builds an interest profile (creator, platform, keywords, length, shape); old interests fade over 14 days.
- Ranks videos by what you are likely to watch, like YouTube's watch-time ranking.
- About one video in five explores something new, so the feed does not narrow to one topic.
- Holds back repeats of one creator or subject; good videos seed searches for more like them.
- Everything stays in one file on the phone, and it learns only while switched on.

SanViral - the best videos first

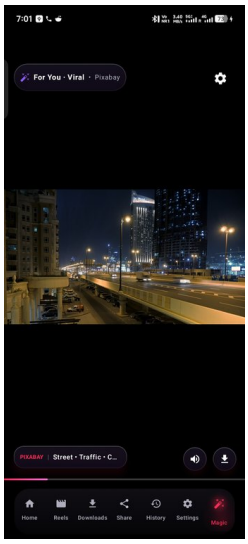
- Reads what each platform publishes: views, views per day, likes, shares and comments per view.
- Ranks each video against others from the same platform, so small platforms are not buried.
- Shows the same clip once, the copy that did best.

My websites - fast grab

- Reads a site's full video links without playing anything, like a download manager.
- Shares each page fairly between sites (3 videos per site, 12 per page); first results within 8 s.
- A site you switch off is never loaded.



Magic feed with Download button



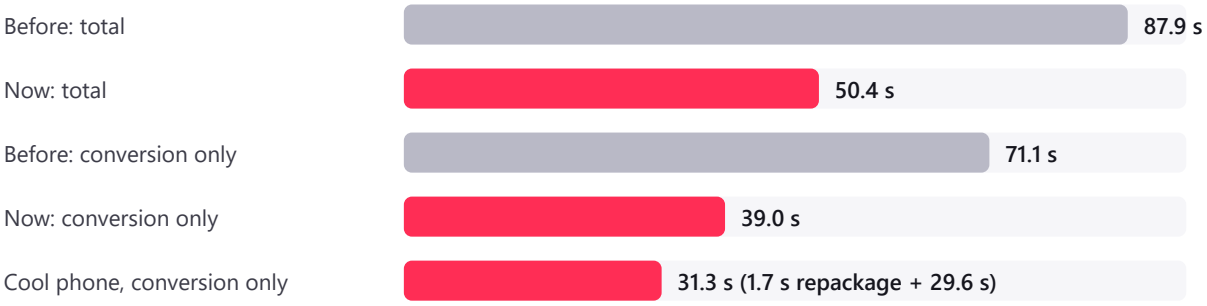
“For You · Viral”

6 · PERFORMANCE

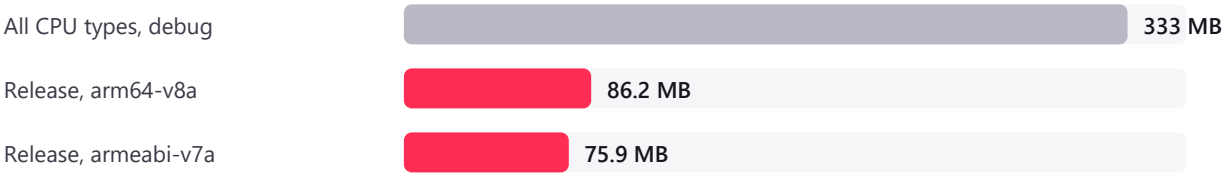
Measured on real phones

All numbers were measured on SanTube's own test phones - a vivo I2302 (MediaTek MT6886, 8 cores) and a vivo V2443 - with real YouTube downloads, not in an emulator.

MP3 download of a 1 h 35 min YouTube track (seconds)



APK size (MB)



Measurement	Result
MP3: download / convert / tags, before → now	8.7 / 71.1 / 5.5 s → 7.4 / 39.0 / 2.4 s
Analyzing a YouTube link	about 4-5 s for all formats; the title appears after about 0.8 s
YouTube download, yt-dlp's own downloader	a 27 MB video in 1.5 s (parallel connections are slowed to ~1 MiB/s)
Other sites, aria2c with 16 connections	100 MB in 19 s instead of about 3 minutes
LAME effort level 7 instead of the default	about 7 s instead of 12 s per 10 minutes of audio, same bitrate
Engine start when the app opens again	about 20-30 ms (already unpacked)

7 · WHAT WAS DONE IN 1.1

Problems found, and how they were fixed

Version 1.1 came from testing on real phones and from friends' feedback. Each fix was checked on a phone; several were designed after studying how Snaptube and VLC do the same thing, using the REA analysis tool.

Problem	What changed	Checked
Instagram links failed on a friend's fresh install	Newer yt-dlp in the APK; updates also when opened from a share; update and retry after a failure	fresh-install test: video found
Each shared link opened another SanTube window	SanTube keeps one window; new links go to it	Recents shows one SanTube
Back closed the whole app	Back goes to Home first, then leaves	every tab tested
Bottom bar too small, fixed order	Size 80-170 %, tab order, show/hide (like VLC's “Rearrange order”)	kept after reinstall
No picture in the finished notification	Video thumbnail and full title, as Snaptube does	90×90 thumbnail shown
MP3 conversion slow	Parallel, seamless MP3 encoder	87.9 s → 50.4 s, byte-identical
Download button hidden in the floating window	List shrinks and scrolls, button pinned (Snaptube's layout)	visible in a 400 dp window
APK 333 MB to share	One APK per CPU type, R8 shrinking	86 MB, installs as an update
Downloads stuck at 0 % in the release build	Keep rules for WorkManager after R8 removed a constructor	download completes
Share, History and the download sheet were basic	Snaptube-style Share list, detailed History, “Download video as” sheet, copied-link card	tested on both phones
Magic “My websites” slow or loading switched-off sites	Fast grab, fair sharing between sites, switched-off sites skipped	each site tested alone
Yellow buttons did not match the brand	Red-and-white neon theme with glass tiles	-

8 · QUALITY, LIMITS AND TECH

How it is tested, and what it cannot do

Quality

- **86 automated tests** for the logic that matters: the format list, MP3 frame joining, the bottom bar's order rules, SanAlgorithm and SanViral ranking, progress parsing, saved websites.
- **Real-phone testing** on two vivo phones for every change: downloads, notifications, background play, share handling, the floating window.
- **Clean source**: the GitHub copy holds only the source (about 8 MB); it builds straight from a downloaded ZIP.

Drawbacks and limits

- **Large app**: Python, FFmpeg and libVLC make about 86 MB per CPU type.
- **Sites change**: downloading works as long as yt-dlp understands the site; SanTube updates it by itself, but a site can fail for a few days until yt-dlp catches up.
- **No login**: private, age-restricted or members-only videos fail; Instagram may refuse a phone that downloads a lot.
- **MP3 takes time**: much faster than before, but not instant; Fast M4A needs no conversion at all.
- **Not for Google Play**: Play does not allow YouTube downloaders, and builds are signed with a debug key.
- **Tested mostly on two vivo phones**; the 32-bit build and other brands are less tested.
- **Use responsibly**: download only what you have the right to, and respect creators' copyright.

Tech stack

Area	Used
Language and UI	Kotlin 2.2, Jetpack Compose (BOM 2026.02.01), Material 3
Download engine	youtubedl-android 0.18.1 (yt-dlp + Python), FFmpeg 7.1 with LAME, aria2c
Playback	AndroidX Media3 1.4.1 (ExoPlayer, MediaSession), libVLC 3.7.7
Background and storage	WorkManager 2.9, Room 2.7, MediaStore, SharedPreferences
Network and images	OkHttp, Coil
Build	Android Gradle Plugin 9.2.1, Gradle 9.6, R8, Java 21 - minSdk 24, targetSdk 36

Developed by **SanStudio** · **Santhosh**. Built on yt-dlp, youtubedl-android, FFmpeg, LAME, aria2c and libVLC.